Software Requirements Specification

Only for course Teacher						
		Needs Improvement	**Developing**	**Sufficient**	**Above Average**	**Total Mark**
Allocate mark & Percentage		**25%**	**50%**	**75%**	**100%**	**15**
Clarity	**3**					
Content Quality	**6**					
Spelling & Grammar	**3**					
Organization and Formatting	**3**					
					Total obtained mark	
Comments						

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BlindGuide

1. Introduction

BlindGuide is an AI-powered assistive application designed to help blind and visually impaired people understand their surroundings and move more independently. The system will use a smartphone camera, Artificial Intelligence (AI), Computer Vision, Optical Character Recognition (OCR), voice interaction, and location services to provide real-time environmental information and safety guidance.

The application will detect common objects, identify potential obstacles, estimate the proximity of nearby objects, describe the surrounding environment, read visible text, and communicate relevant information through voice output. It will also provide basic GPS-based navigation and an emergency alert feature.

The primary goal of BlindGuide is to improve accessibility and support independent daily activities through intelligent visual assistance using commonly available smartphone technology.

1.1 Problem Statement

Blind and visually impaired people often face difficulties when navigating unfamiliar environments, identifying nearby objects, detecting obstacles, reading signboards, recognizing room numbers, and understanding surrounding situations.

Traditional walking sticks can help detect certain physical obstacles, but they cannot explain what an obstacle is or describe the overall environment. Conventional camera applications can capture images but cannot independently interpret visual information and provide meaningful spoken guidance.

These limitations can make everyday activities more challenging, particularly in unfamiliar buildings, public places, educational institutions, and outdoor environments.

Artificial Intelligence is needed to analyze visual information, recognize objects, interpret environmental context, extract text, estimate object proximity, and prioritize important safety information. Without AI-based analysis, the application would not be able to provide intelligent, context-aware environmental assistance.

Therefore, BlindGuide proposes an integrated AI-based solution that transforms smartphone camera input into understandable voice instructions to support blind and visually impaired users.

1.2 Purpose

The purpose of BlindGuide is to develop an accessible mobile application that uses AI to interpret the user's surroundings and provide useful environmental information through voice assistance.

The primary objectives are:

1. To detect and recognize common objects using Computer Vision.
2. To identify potential obstacles in the user's walking path.
3. To estimate the approximate proximity of detected objects.
4. To generate meaningful descriptions of surrounding environments.
5. To recognize and read visible text using OCR.
6. To provide voice-based interaction and spoken environmental guidance.
7. To support basic GPS-based navigation.
8. To provide an emergency alert mechanism for sharing the user's location with configured contacts.
9. To improve accessibility through a simple, voice-oriented mobile interface.
10. To evaluate the AI components and overall system through functional, performance, and usability testing.

1.3 Scope

In-Scope

1. AI Object Detection: The system will use smartphone camera input to recognize common objects such as people, chairs, doors, vehicles, bicycles, stairs, and other supported object categories.

2. Obstacle Detection: The system will analyze detected objects and available visual information to identify potential obstacles that may affect the user's movement.

3. Proximity Estimation: The system will estimate the approximate proximity of detected objects and classify them into categories such as Near, Medium, and Far.

4. AI Scene Understanding: The system will analyze the visual context and generate simple descriptions of the surrounding environment.

5. OCR and Text Recognition: The application will extract visible text from supported sources such as signboards, room numbers, labels, notices, and documents.

6. Voice-Based Assistance: Users will be able to interact with the application using supported voice commands and receive spoken responses.

7. Navigation Assistance: The application will provide basic GPS-based navigation and directional information using an appropriate map or navigation service.

8. Emergency Alert: Users will be able to trigger an emergency alert and share their available location information with configured emergency contacts.

9. Accessible Mobile Interface: The application will support voice output, clear audio feedback, accessible controls, and simple interaction suitable for blind and visually impaired users.

2. Design and Implementation Constraints

This section defines the proposed technology stack and the technical boundaries of the project. The final selection of libraries, AI models, and services will depend on feasibility testing, available resources, licensing, and performance on the target smartphone.

2.1 Programming Language

Python

Python will be used for AI and Computer Vision development because it supports image processing, object detection, model evaluation, and integration with machine-learning libraries.

JavaScript

JavaScript will be used for the mobile application and backend development if React Native is selected. TypeScript may be preferred for improved code organization and type safety.

Dart

Dart will be used if Flutter is selected for mobile application development instead of React Native.

Proposed choice: Python for AI services and JavaScript for the mobile application and backend. The mobile framework will be finalized after a short technical prototype.

2.2 Database and Servers

Database

MySQL or PostgreSQL will be used to store user accounts, accessibility preferences, emergency contacts, and relevant safety-event history.

SQLite or local mobile storage may be used for appropriate device-side settings and cached information.

Backend

Node.js with Express.js will provide REST APIs for user authentication, profile management, event history, and emergency-contact configuration.

A Python-based AI service may be developed separately to handle computer-vision tasks and AI inference.

Data Flow

The mobile application will communicate with the backend through REST APIs when network access is required. The backend may communicate with the AI service to process supported requests.

For privacy and performance, suitable lightweight AI models may run directly on the smartphone where technically feasible. Sensitive camera images should not be uploaded or retained unnecessarily.

Constraints

- AI processing performance will depend on smartphone hardware and the selected model.
- Cloud-based AI services will require internet connectivity and may introduce latency or usage costs.
- GPS accuracy may decrease indoors or in areas with poor satellite reception.
- Emergency alerts will depend on device permissions, network availability, and the configured communication service.
- Database and API access must be protected by appropriate authentication and authorization.

2.3 AI Model and Services

BlindGuide will use a combination of Computer Vision, OCR, scene-understanding techniques, and speech technologies.

AI Component	Proposed Approach	Purpose
Object Detection	YOLO-family model or a compatible lightweight detection model	Identify supported objects in camera input
Image Processing	OpenCV	Prepare images and support visual processing

AI Component	Proposed Approach	Purpose
Obstacle Analysis	Object detection combined with path/context analysis	Identify potential obstacles
Proximity Estimation	Monocular depth estimation or calibrated visual techniques	Estimate approximate object proximity
Scene Understanding	Vision-Language Model (VLM) or a suitable image-captioning model	Generate contextual scene descriptions
OCR	Tesseract OCR or Easy OCR	Extract text from images
Speech Recognition	Android Speech Recognition or a suitable Speech-to-Text service	Receive supported voice commands
Text-to-Speech	Android Text-to-Speech or a suitable TTS engine	Speak AI-generated information
Navigation	GPS and a suitable map/routing service	Provide basic directional guidance

3. User Classes and Characteristics

BlindGuide will support the following user classes.

3.1 Blind and Visually Impaired Users

These are the primary users of the application.

Permissions and activities:

- Access the camera-based assistance features.
- Receive spoken object descriptions and obstacle warnings.
- Request text recognition and scene descriptions.
- Interact through supported voice commands.
- Use basic GPS navigation.
- Configure accessibility preferences and emergency contacts.
- Trigger emergency alerts.
- View or request their own available assistance history.

3.2 Caregivers and Emergency Contacts

Caregivers or trusted contacts may receive emergency notifications when the user activates the emergency feature.

Permissions and activities:

- Receive emergency alerts from the user.
- View the shared location information included in an alert.

- Contact the user when necessary.

Caregivers will not automatically receive continuous camera access or live location tracking. Location sharing will be limited to the functionality and permissions implemented in the system.

3.3 System Administrator

The administrator will maintain the technical operation of the application and backend.

Permissions and activities:

- Manage authorized user accounts where required.
- Monitor backend and service availability.
- Maintain system configuration and security settings.
- Review technical error logs.
- Manage supported AI service configurations.

The administrator will not have unrestricted access to private camera content or personal information unless access is necessary, explicitly authorized, and appropriately protected.

3.4 Development and Testing Team

The development team will build, integrate, test, and maintain the system.

Responsibilities:

- Develop the mobile application and backend.
- Integrate and evaluate AI models.
- Test accessibility, performance, and security.
- Maintain documentation and deployment configurations.

4. Requirement Elicitation

Requirement elicitation is the process of identifying and analyzing the needs of the intended users and translating those needs into software requirements. Since BlindGuide is designed for blind and visually impaired people, the team needs to understand the challenges they may face when identifying objects, navigating environments, reading text, and receiving safety information.

As the team could not directly access blind or visually impaired participants, hospitals, or relevant NGOs during the initial research stage, secondary research and document analysis were used as the preliminary requirement elicitation method. The team

reviewed published research papers, systematic reviews, and official documentation of existing assistive technologies.

The findings from these sources were analyzed to identify relevant system features, technical challenges, and potential requirements for BlindGuide. These findings represent evidence from existing literature and products, not direct feedback collected from the project's target users.

4.1 Requirement Elicitation Techniques

4.1.1 Secondary Research

Secondary research was selected as the primary requirement elicitation method for the initial version of BlindGuide. It involved reviewing existing research and documentation related to AI-based assistive technologies for blind and visually impaired people.

The research focused on the following areas:

- Smartphone-based assistive technology.
- AI-powered object recognition and obstacle detection.
- Scene understanding and environmental description.
- Text recognition and text-to-speech.
- Approximate object proximity and navigation assistance.
- Voice-based feedback and accessibility.
- Usability, reliability, privacy, and limitations of AI-based assistance.

4.1.2 Document Analysis

The team examined academic papers, systematic reviews, and official product documentation to identify common features and limitations of existing assistive solutions.

The following sources were selected because they are directly related to the proposed project:

Source 1: Smartphone-Based Computer Vision Travelling Aids

- **Title:** *Smartphone-based computer vision travelling aids for blind and visually impaired individuals: A systematic review*
- **Authors:** Andrius Budrionis, Darius Plikynas, Povilas Daniušis, and Audrius Indrulionis.
- **Published:** 2020; journal issue published in 2022.
- **Source:** PubMed.
- **URL:** https://pubmed.ncbi.nlm.nih.gov/32207640/

This systematic review examined smartphone-based assistive solutions that use computer vision to support orientation and navigation in indoor and outdoor environments. It discussed the use of surrounding-object information, object recognition, and obstacle detection.

Requirement implications: BlindGuide should consider smartphone-based visual assistance, object detection, obstacle analysis, and navigation-related information. The paper also highlights a mismatch between some research developments and the needs of blind users, showing the importance of validating the system with actual users when possible.

Source 2: Computer Vision-Driven Scene Understanding

- **Title:** *Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping*
- **Authors:** Mohammad Moeen Valipoor and Angélica de Antonio.
- **Published:** Online in 2022; journal issue published in 2023.
- **Source:** Springer Nature.
- **URL:** https://link.springer.com/article/10.1007/s10209-022-00868-w

This systematic mapping reviewed research on computer-vision-based assistance for blind and visually impaired users. The study covered object recognition, obstacle detection, scene understanding, and related assistive services. It also discussed usability and technical challenges in existing approaches.

Requirement implications: BlindGuide should include object detection, potential-obstacle analysis, scene descriptions, and understandable audio feedback. Proximity estimation should be treated as approximate, and the system must be evaluated for usability, latency, and reliability.

Source 3: Google Lookout

- **Title:** *Use Lookout to explore your surroundings*
- **Organization:** Google Accessibility Help.
- **Source:** Official Google documentation.
- **URL:** https://support.google.com/accessibility/android/answer/9031274?hl=en

Google Lookout is an existing Android application designed to help blind and low-vision users obtain information about their surroundings. Its documented features include recognizing objects and text, reading documents, describing images, and providing information about selected objects.

Requirement implications: The documented features support including camera-based object recognition, OCR, text-to-speech, and scene-description functionality in BlindGuide. The documentation also notes that image quality, lighting, and other conditions can affect results, so the proposed system should handle unsuccessful or uncertain AI processing clearly.

Source 4: Microsoft Seeing AI

- **Title:** *Seeing AI App Launches on Android—Including New and Updated Features and New Languages*
- **Organization:** Microsoft Accessibility.
- **Published:** December 4, 2023.
- **URL:** https://blogs.microsoft.com/accessibility/seeing-ai-app-launches-on-android-including-new-and-updated-features-and-new-languages/

Microsoft Seeing AI is an assistive application that uses AI to help blind and low-vision users obtain spoken information about text, products, people, and images. Its documented features include reading printed text, describing scenes, and providing information about selected visual content.

Requirement implications: The documented features support text recognition, scene description, and spoken AI feedback as relevant capabilities for BlindGuide. They also demonstrate that the proposed feature set is technically related to an existing category of assistive applications. BlindGuide must therefore be evaluated on its own implementation and usefulness rather than assuming that its planned features are automatically novel or superior.

4.1.3 Evidence Review Procedure

The team followed these steps during the preliminary document analysis:

1. Identified research papers and official documentation related to AI-based assistance for blind and visually impaired people.
2. Reviewed the available titles, abstracts, descriptions, and relevant feature information.
3. Identified recurring technical capabilities, including object recognition, obstacle detection, text reading, scene description, and audio feedback.
4. Recorded the limitations discussed in the sources, such as usability challenges, image-quality dependence, and uncertainty in visual interpretation.
5. Mapped the relevant findings to the proposed functional and non-functional requirements of BlindGuide.
6. Prepared a preliminary requirement list for implementation and future validation.

4.2 Elicitation Evidence

The evidence for the initial requirement analysis consists of published research papers and official documentation of existing assistive technologies. The sources provide evidence that the selected features are relevant to the field of AI-based assistive technology.

The following table summarizes the reviewed evidence.

Evidence ID	Source	Evidence Identified	Related Requirements
E-01	Budrionis et al., *Smartphone-based computer vision travelling aids for blind and visually impaired individuals: A systematic review*	Smartphone-based visual assistance, object information, orientation, navigation, and obstacle detection are established research areas.	FR-04, FR-05, FR-13
E-02	Valipoor and de Antonio, *Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping*	Object recognition, obstacle detection, and scene understanding are important areas of computer-vision-based assistance. Usability and technical challenges remain relevant.	FR-04, FR-05, FR-07, FR-11, NFR-01, NFR-03, NFR-08
E-03	Google Lookout official documentation	Existing assistive functionality includes object and text recognition, document reading, image description, and information about selected objects.	FR-04, FR-07, FR-08, FR-09
E-04	Microsoft Seeing AI official documentation	Existing AI-based assistance includes spoken text reading, scene descriptions, product information, and other visual-information features.	FR-07, FR-08, FR-09, FR-10

The team will retain the source URLs and available article copies or permitted excerpts in the project appendix. Each source should be identifiable by its title, authors or organization, publication date where available, URL, and access date.

The reviewed sources support the relevance of the proposed feature categories. However, they do not establish that BlindGuide already achieves any particular accuracy, response time, usability level, or safety performance. Those outcomes must be measured through implementation and testing.

4.3 Key Findings

Since direct interviews, questionnaires, and observations were not conducted, the key findings for BlindGuard were identified through secondary research, including published research papers, systematic reviews, and existing assistive applications.

The reviewed sources indicate the following findings:

1. **Object Recognition and Scene Understanding:** Research on computer vision-based assistive systems identifies object recognition and scene understanding as important capabilities for helping blind and visually impaired people obtain information about their surroundings.
2. **Obstacle Detection and Distance Awareness:** Research highlights obstacle detection and depth estimation as important areas of assistive technology. However, accurately estimating the distance and position of real-world objects remains challenging.
3. **Text Recognition:** Existing applications such as Google Lookout provide camera-based text and document reading, demonstrating the relevance of OCR technology for accessing printed information.
4. **Audio-Based Assistance:** Existing assistive applications communicate detected information through spoken output, supporting the use of text-to-speech in BlindGuard.
5. **Navigation Assistance:** Published research explores smartphone-based computer vision systems for orientation, navigation, and wayfinding. These findings support investigating navigation assistance as part of BlindGuard, while recognizing the limitations of camera-based perception and location accuracy.

Based on these findings, BlindGuard will focus on AI-powered object detection, obstacle identification, proximity estimation, scene understanding, text recognition, and voice-based assistance. Navigation support and emergency alerts will complement these core features.

4.4 Requirement Changes After Elicitation

Direct user elicitation was not conducted for this project. Therefore, the following table documents how the initial requirements were refined through secondary research and analysis of existing assistive applications. The changes are research-informed design decisions rather than changes resulting from interviews or questionnaires.

ID	Change	Before Secondary Research	After Secondary Research	Source / Reason
FR-1	Modified	Identify objects using the smartphone camera.	The system shall identify relevant objects and provide information about the surrounding scene.	Research on computer vision-based assistive systems identifies object recognition and scene

				understanding as key capabilities.
FR-2	Added	No separate obstacle analysis requirement.	The system shall identify potential obstacles and provide audio warnings when relevant.	Obstacle detection is identified as an important feature in assistive scene-understanding research.
FR-3	Added	No explicit proximity estimation requirement.	The system shall estimate an object's proximity when supported by the available vision model and device capabilities.	Research identifies depth and distance estimation as challenging but relevant capabilities for assistive systems.
FR-4	Added	No dedicated text-reading requirement.	The system shall recognize visible text using OCR and read the recognized text aloud.	Google Lookout provides text and document reading through its camera-based interface.
FR-5	Modified	Provide visual information to the user.	The system shall communicate important AI-generated information through speech output.	Existing assistive applications use spoken feedback to make detected information accessible.
FR-6	Added	No explicit navigation assistance requirement.	The system shall provide basic navigation assistance using available location and direction information.	Systematic research reviews smartphone-based assistive technologies for orientation and navigation.
1FR-7	Added	No explicit emergency alert requirement.	The system shall allow the user to trigger an emergency alert to configured contacts, subject to available device permissions and connectivity.	This is a proposed safety-support feature of BlindGuard; it is not claimed as a finding directly established by the reviewed sources.

Research sources:

1. Budrionis et al., *Smartphone-based computer vision travelling aids for blind and visually impaired individuals: A systematic review*. PubMed

2. Valipoor and de Antonio, *Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping*. Springer
3. Google, *Use Lookout to explore your surroundings*. Official documentation

References

1. Budrionis, A., Plikynas, D., Daniušis, P., & Indrulionis, A. (2020; journal issue 2022). *Smartphone-based computer vision travelling aids for blind and visually impaired individuals: A systematic review*. PubMed. https://pubmed.ncbi.nlm.nih.gov/32207640/
2. Valipoor, M. M., & de Antonio, A. (2022; journal issue 2023). *Recent trends in computer vision-driven scene understanding for VI/blind users: a systematic mapping*. Universal Access in the Information Society, Springer Nature. https://link.springer.com/article/10.1007/s10209-022-00868-w
3. Google. *Use Lookout to explore your surroundings*. Google Accessibility Help. https://support.google.com/accessibility/android/answer/9031274?hl=en
4. Microsoft Accessibility. (2023, December 4). *Seeing AI App Launches on Android – Including new and updated features and new languages*. Microsoft Blogs. https://blogs.microsoft.com/accessibility/seeing-ai-app-launches-on-android-including-new-and-updated-features-and-new-languages/

5. Functional Requirements (FR)

The functional requirements define the core capabilities of BlindGuide. Priorities are proposed for the initial prototype. Assignees refer to the planned team distribution and may be updated as the project develops.

ID	Requirement Name	Description	Users	Priority	Assignee
FR-01	User Registration and Login	The system shall allow users to register and securely log in to the application.	Primary User, Admin	Medium	Maruf
FR-02	User Profile Management	The system shall allow users to manage accessibility preferences and relevant profile information.	Primary User	Medium	Maruf
FR-03	Camera Integration	The system shall access the smartphone camera after obtaining the	Primary User	High	Saki

		required permissions and capture visual input for assistance.			
FR-04	AI Object Detection	The system shall detect and identify supported objects in camera input and provide their recognized labels.	Primary User	High	Mahin
FR-05	Obstacle Detection	The system shall analyze detected objects and visual context to identify potential obstacles in the user's path.	Primary User	High	Mahin
FR-06	Proximity Estimation	The system shall estimate the approximate proximity of supported objects and classify them as Near, Medium, or Far when sufficient information is available.	Primary User	High	Mahin
FR-07	AI Scene Understanding	The system shall analyze a visual scene and generate a concise description of relevant surroundings.	Primary User	High	Mahin
FR-08	OCR and Text Recognition	The system shall extract visible text from supported images, including signboards, room numbers, labels, and notices.	Primary User	High	Mahin
FR-09	Text-to-Speech	The system shall convert supported recognized text and AI-generated information into spoken output.	Primary User	High	Saki
FR-10	Voice-Based Assistance	The system shall accept supported voice commands or	Primary User	High	Saki

		questions and provide relevant responses.			
FR-11	AI Safety Analysis	The system shall combine available object, obstacle, proximity, and scene information to identify potential safety concerns.	Primary User	High	Mahin
FR-12	Real-Time Voice Warning	The system shall provide a spoken warning when the system detects a supported potential hazard that meets the configured warning criteria.	Primary User	High	Mahin, Saki
FR-13	GPS Navigation Assistance	The system shall provide basic directional guidance to a selected destination using an available navigation service.	Primary User	Medium	Saki
FR-14	Emergency Alert	The system shall allow the user to trigger an emergency alert and share available location information with configured emergency contacts.	Primary User, Emergency Contact	High	Maruf
FR-15	Safety Event History	The system shall record relevant events, such as warnings and emergency-alert status, and allow authorized access to the available history.	Primary User, Admin	Medium	Maruf
FR-16	Accessibility Settings	The system shall allow users to configure supported voice, language, and assistance preferences.	Primary User	Medium	Saki

FR-17	Permission Management	The system shall request and handle camera, microphone, location, and notification permissions as required by the selected features.	Primary User	High	Saki
FR-18	AI Processing Error Handling	The system shall inform the user when an AI feature fails, becomes unavailable, or cannot confidently interpret the input.	Primary User	High	Mahin, Saki

5.1 Requirement Priorities

- **High:** Essential for the core prototype or its safety-related operation.
- **Medium:** Important supporting functionality that may be implemented after the core AI workflow.

These priorities are initial planning decisions and may change after requirement elicitation and feasibility testing.

6. Non-Functional Requirements

Non-functional requirements define the quality, performance, security, usability, and reliability expectations of BlindGuide.

The numerical targets below are proposed engineering targets for testing. They are not claims of achieved performance and must be validated on the selected devices and models.

6.1 Performance

- The object-detection and warning pipeline should target a response time of no more than 2 seconds under defined test conditions on a supported device.
- The application should minimize unnecessary camera processing and background activity.
- Voice warnings should be generated promptly after a relevant hazard is detected.

6.2 Security

- Passwords must not be stored in plaintext.
- Authenticated APIs must enforce authorization and secure communication.
- Emergency contacts, account information, and stored location data must be protected against unauthorized access.
- Administrative access must be restricted to authorized personnel.

6.3 Usability and Accessibility

- The application must provide a voice-oriented interaction experience.
- Important controls must be accessible to Android screen readers where applicable.
- The interface must use understandable labels, consistent navigation, and clear error messages.
- The system should allow users to stop or pause assistance when needed.
- Warnings should be concise and should avoid unnecessary repetition.

6.4 AI Quality

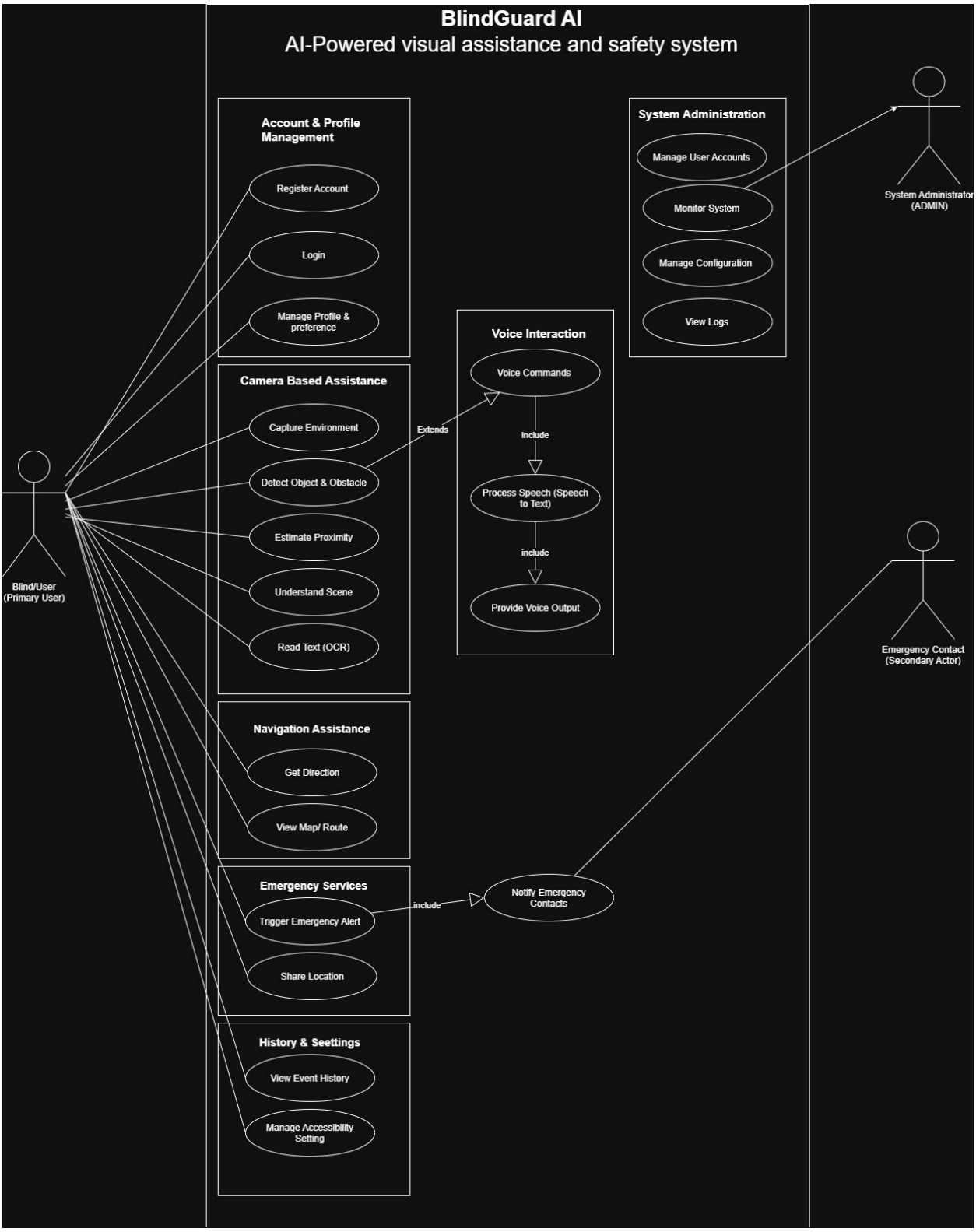
- The object-detection component should be evaluated using precision, recall, and an appropriate detection metric such as MAP.
- OCR should be evaluated using character or word recognition accuracy on a representative test set.
- Warning latency should be measured under defined conditions.
- False warnings and missed hazards should be recorded during controlled tests.
- The system must communicate uncertainty when model output is unreliable.
- No minimum accuracy figure will be reported as achieved until actual evaluation has been completed.

7. System Design Diagrams

This section describes the required diagrams and the core design logic of BlindGuide.

7.1 Use Case Diagram

The following diagram shows the main actors and their interactions with the system.



Description: The primary user interacts with the mobile application through camera-based assistance, voice commands, text recognition, navigation, and emergency

functions. Emergency contacts receive alerts when the user triggers the relevant feature. Administrators manage technical configuration and system operation.

7.2 Use Case Descriptions

The following descriptions cover key high-priority functions.

Use Case 1: Detect Objects and Warn About Obstacles

Use Case ID: UC-01
Primary Actor: Blind or Visually Impaired User
Priority: High

Preconditions:

- The application is installed and running.
- Camera permission has been granted.
- The selected AI model is available.

Main Success Flow:

1. The user starts camera-based assistance.
2. The application captures a frame from the smartphone camera.
3. The AI model detects supported objects.
4. The system analyzes relevant detections and available visual context.
5. The system estimates proximity when the selected technique supports it.
6. The safety component evaluates whether a warning is appropriate.
7. The application generates a concise spoken warning or object description.
8. The user receives the audio feedback.

Alternative Flows:

- If camera permission is denied, the application explains how to enable it.
- If no relevant object is detected, the system does not invent an object description.
- If the model fails, the application informs the user that visual assistance is temporarily unavailable.
- If confidence is low, the system may report uncertainty instead of issuing a definitive warning.

Postcondition: A relevant result or an understandable error message is provided to the user.

Use Case 2: Read Visible Text

Use Case ID: UC-02
Primary Actor: Blind or Visually Impaired User
Priority: High

Preconditions: The camera is available, and the OCR component is accessible.

Main Success Flow:

1. The user selects the text-reading feature or issues a supported voice command.
2. The application captures an image of the text.
3. The image is prepared for OCR processing.
4. The OCR component extracts readable text.
5. The application checks whether usable text was returned.
6. The text is converted into speech.
7. The user listens to the recognized content.

Alternative Flows:

- If the text is blurred or unreadable, the application requests a clearer image.
- If no text is detected, the application informs the user.
- If the voice service fails, the application reports the error and offers another supported way to access the text.

Postcondition: Recognized text is spoken to the user, or the user receives a clear failure message.

Use Case 3: Trigger an Emergency Alert

Use Case ID: UC-03
Primary Actor: Blind or Visually Impaired User
Secondary Actor: Configured Emergency Contact
Priority: High

Preconditions:

- At least one emergency contact has been configured.
- Required communication permissions or services are available.
- Location permission has been granted if location sharing is requested.

Main Success Flow:

1. The user activates the emergency-alert function.
2. The application confirms the user's request when required by the implemented interaction design.

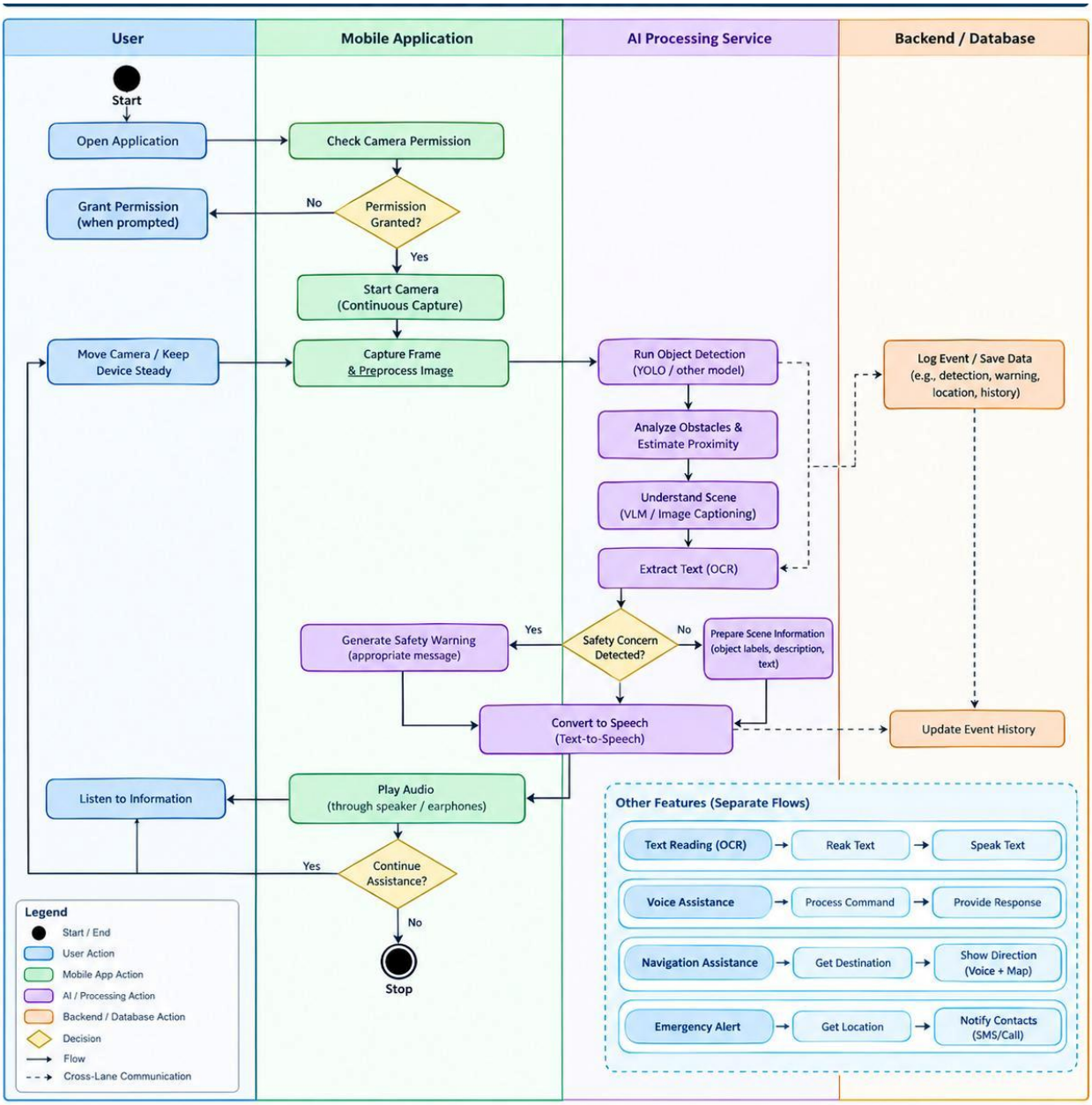
3. The system retrieves available location information if permission allows.
4. The application prepares an emergency message.
5. The communication service attempts to send the message to the configured contact.
6. The application reports the available delivery status to the user.
7. The system records the event and its status where supported.

Alternative Flows:

- If location is unavailable, the system informs the user and follows the configured fallback behavior.
- If the communication service is unavailable, the application reports that the alert could not be confirmed as delivered.
- If no contact is configured, the application asks the user to configure an emergency contact.

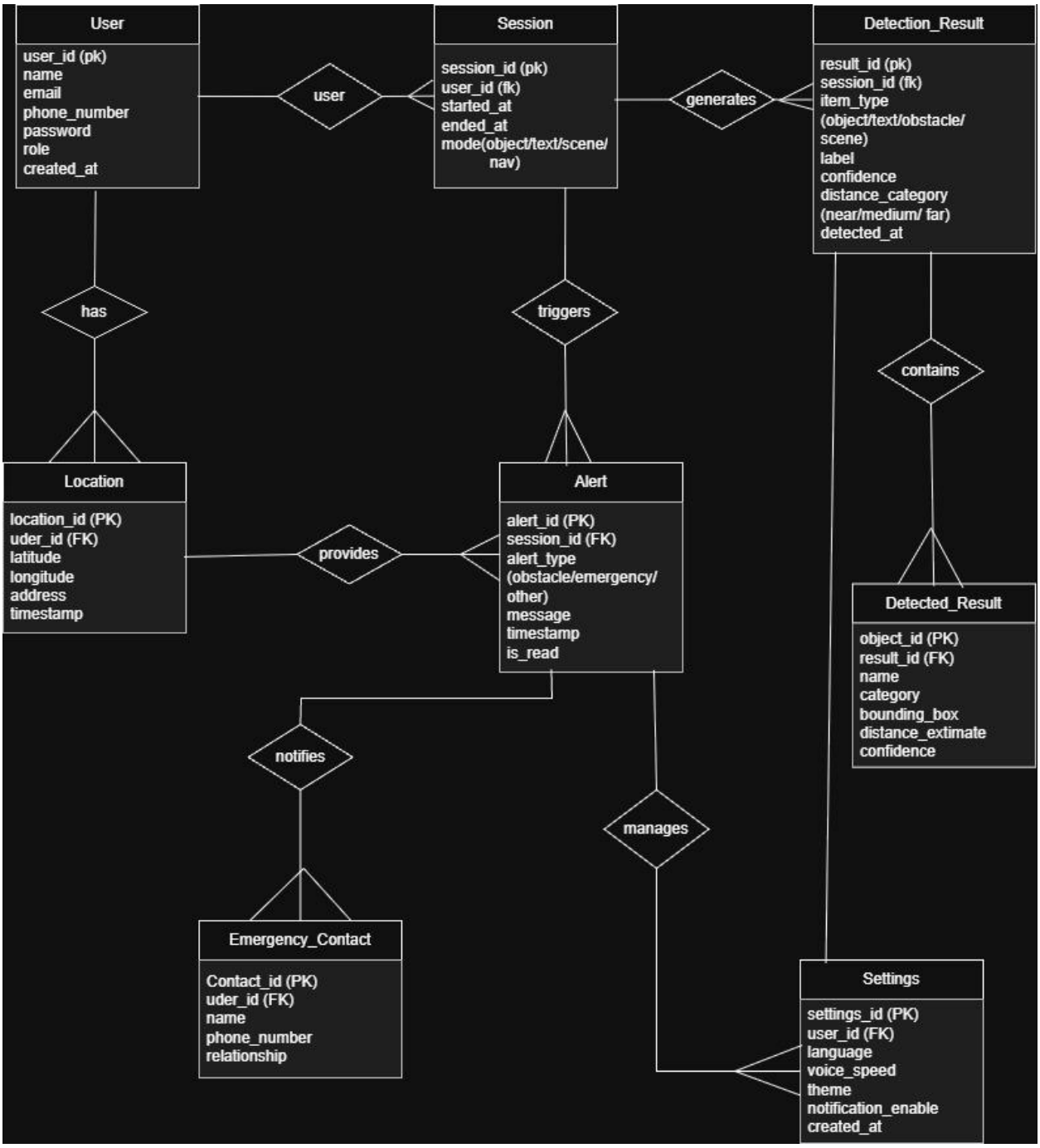
Postcondition: The alert attempt and available status are recorded, and the application does not falsely claim successful delivery.

7.3 Activity Diagram



Description: The application checks camera permission, captures visual input, processes the image, and runs the AI components. It then decides whether a safety warning or general environmental description is more appropriate. The result is communicated through voice, and the process continues while assistance remains active.

7.4 Entity Relationship Diagram (ERD)



Entity Relationships

- One user may generate many safety events (1).
- One user may configure multiple emergency contacts (1).
- One user may trigger multiple emergency alerts (1).
- One emergency contact may receive multiple emergency alerts over time (1).

- One user may have one preference record in the proposed design (1:1).
- One safety event may have multiple AI analysis records (1).

The database should avoid retaining raw camera images by default. The detected_objects field may be implemented as structured JSON or a related table, depending on the selected database design. Precise location should be stored only when required by the feature and covered by the privacy design.

7.5 AI Feature Design

A. Data

The AI components will use visual input captured through the smartphone camera. Depending on the chosen model, development may also use publicly available datasets, appropriately licensed pretrained models, and a small project-specific evaluation dataset.

Proposed data sources:

- Public object-detection datasets containing supported everyday object categories.
- Public OCR datasets or representative text images.
- Publicly available scene-understanding datasets or pretrained models.
- Project-specific test images and video clips captured with appropriate consent.

Data Preparation:

1. Select data relevant to supported object categories and intended environments.
2. Review dataset licenses and permitted usage.
3. Resize or normalize images according to model requirements.
4. Apply suitable preprocessing.
5. Prepare training, validation, and test splits when model training is necessary.
6. Label or verify test examples as required.
7. Remove identifying information from project-specific samples where possible.

The dataset size, label distribution, and final train/validation/test split will be documented after selection. No dataset size or training result is assumed in advance.

B. Model

The proposed AI approach combines pretrained computer-vision models with supporting processing modules.

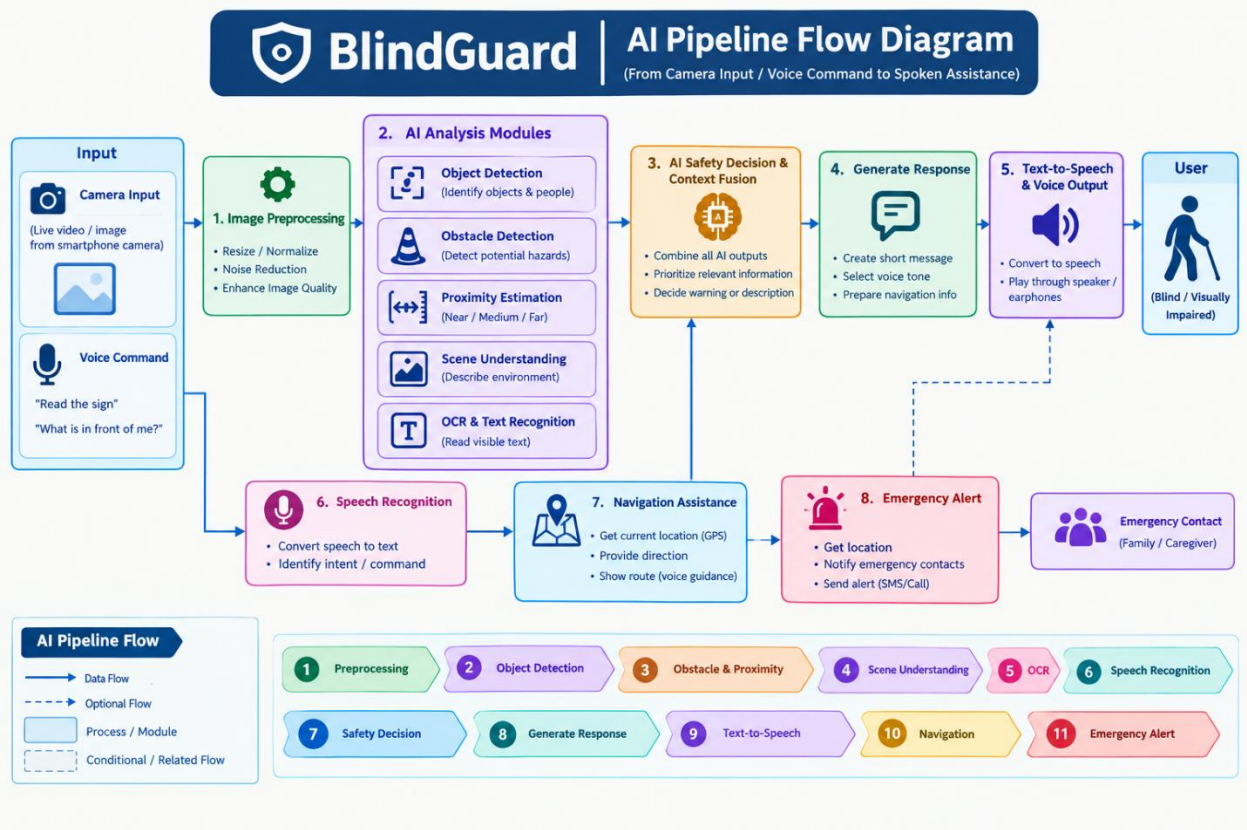
Component	Proposed Model or Method	Reason
Object Detection	YOLO-family or another lightweight detector	Detect supported everyday objects
Image Processing	OpenCV	Support preprocessing and image operations
Proximity Estimation	Depth estimation or calibrated visual estimation	Provide approximate proximity categories
Scene Understanding	Suitable VLM or image-captioning model	Describe relevant environmental context
OCR	Tesseract OCR or Easy OCR	Extract visible text
Voice Input	Android Speech Recognition or suitable API	Process supported voice commands
Voice Output	Android Text-to-Speech or suitable TTS engine	Speak results

Baseline comparison:

- Compare the selected object detector with a simple baseline or alternative compatible model.
- Compare OCR results against manually verified text on a held-out test set.
- Compare AI-based warning decisions with a predefined rule-based baseline.
- Compare end-to-end response time across representative test conditions.

A more complex model will only be retained if it provides sufficient practical benefit relative to latency, resource use, privacy, and implementation effort.

C. AI Pipeline



Pipeline Explanation:

1. **Input:** The application captures visual input from the smartphone camera or receives a supported voice command.
2. **Preprocessing:** The system prepares the image for the selected AI model.
3. **AI Inference:** The object detector identifies supported objects; other components estimate proximity, interpret scene context, or recognize text.
4. **Safety Decision:** The system evaluates available outputs and prioritizes relevant warnings.
5. **Output Generation:** The selected result is converted into a short spoken message.
6. **User Feedback:** The application provides the message through the device speaker or connected earphones.

D. Integration

The mobile application will coordinate camera access, AI processing, voice output, navigation, and emergency functions.

Depending on the final implementation:

- On-device models may process images directly within the mobile application.
- A Python AI service may expose endpoints for models that run on the backend.
- The Node.js backend may manage authentication, preferences, contacts, and event history.
- REST APIs may transfer supported requests and results between the mobile application and backend.
- AI results may be stored as event metadata when required.

The system should avoid sending every camera frame to a remote server. If a cloud-based model is selected, the application must handle network failure, request latency, privacy, and service cost.

E. Evaluation and Fallback

AI evaluation will use representative test images, videos, and controlled scenarios.

Evaluation Area	Proposed Metric or Method
Object Detection	Precision, recall, MAP
OCR	Character or word recognition accuracy
Scene Understanding	Human review for relevance and factual consistency
Proximity Estimation	Category accuracy against a suitable reference setup
Safety Warnings	False-warning rate, missed-warning rate, and controlled scenario review
Response Time	End-to-end latency measurements
Usability	Task completion, participant feedback, and accessibility checks

Fallback behavior:

- If object detection fails, the application will report that visual assistance is unavailable.
- If scene understanding fails, the application may still provide supported object-detection results.
- If OCR fails, the application will request a clearer image or report that text could not be recognized.
- If GPS is unavailable, navigation will be marked unavailable or limited according to the chosen service.
- If an emergency message cannot be confirmed as delivered, the application will communicate that uncertainty to the user.
- The system will not present unverified model output as a guaranteed safety assessment.

8. Team Member Responsibility Matrix

The team responsibility matrix identifies each member's primary development tasks. Student IDs must be added before final submission.

Member Name	Student ID	Primary Responsibilities
Mahin Al Hasan	241-35-351	AI model selection and evaluation, dataset preparation, object detection, obstacle detection, proximity estimation, scene understanding, AI safety analysis
MD Muhaiminur Rahman Maruf	241-35-052	Backend API development, database schema, authentication, user profiles, emergency-contact management, event history, AI-service integration, backend security
Farhana Jannat Saki	241-35-053	Mobile application development, accessible UI, camera integration, voice interaction, text-to-speech integration, navigation interface, emergency interface, system integration and usability testing

Shared Responsibilities

All three team members will collaborate on:

- Requirement elicitation and documentation.
- System architecture and design diagrams.
- Functional and non-functional testing.
- AI integration and evaluation.
- Privacy and security review.
- Final report preparation, presentation, and demonstration.

Conclusion

BlindGuide is proposed as an AI-powered assistive application designed to help blind and visually impaired people better understand their surroundings through intelligent visual processing and voice-based guidance.

By combining object detection, obstacle analysis, proximity estimation, scene understanding, OCR, voice interaction, navigation, and emergency support, the project aims to address several practical accessibility challenges through a single mobile application.

Artificial Intelligence is central to the project because it enables the application to interpret visual input, identify relevant objects, extract text, understand environmental context, and prioritize useful information for spoken feedback. Without these AI capabilities, the application would not provide the same level of intelligent environmental understanding.

The project will be developed using a modular architecture so that AI models, mobile functionality, backend services, and accessibility features can be tested and improved independently. The initial prototype will focus on a feasible smartphone-based implementation and will be evaluated through controlled tests and, where possible, feedback from blind and visually impaired users.

The system will not be treated as a replacement for a white cane, trained guide, or other established mobility assistance. Instead, it will aim to provide supplementary information while clearly communicating uncertainty and technical limitations.

With further development, BlindGuide may be extended with improved indoor navigation, multilingual assistance, offline AI processing, advanced depth estimation, and compatible wearable devices. These extensions could help make the application a more useful and accessible assistive technology.